



الرسم بالحاسب

9:00-11:00

26/1/2023

ا.د/ خالد فتحي

Faculty of Computers & Information, Assiut University

Course code: (Add the course code)

3rd Level

Final Exam

Duration: 2 hours

Total Marks: 50

Reviewer: (Add the name)

This exam for the following program(s) : (Add the program's name(s))

Hi, Mahmoud. When you submit this form, the owner will see your name and email address.

* Required

1

* الإسم الرباعي (بالعربي فقط)

Enter your answer

2

* رقم الجلوس

Enter your answer

3

* المستوي

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018
- ☐ الرابع 2019

4

* البرنامج

- ☐ بايو
- ☐ هندسة
- ☐ ذكاء صناعي

5

* رقم المعمل

Select your answer



6

* رقم الكمبيوتر

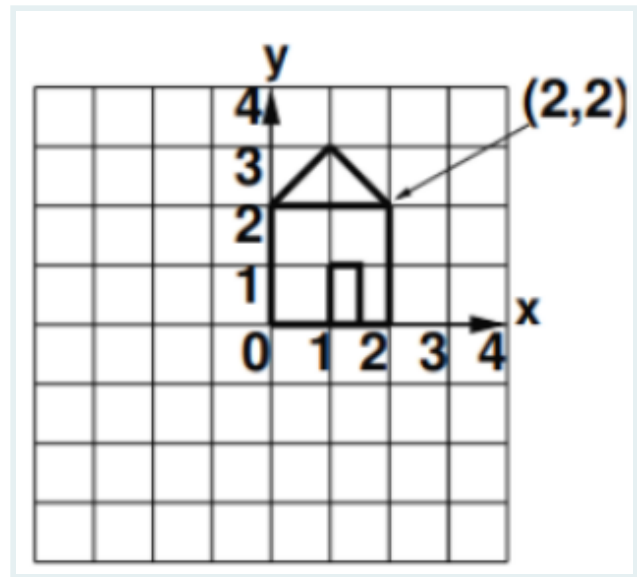
Enter your answer

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

Enter your answer

8



Assume that drawHouse function draws the following house

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef(-1.0, -1.0,1.0);
glRotate(90, 0, 0, 1);
glRotate(90, 0, 0, 1);
drawHouse();
glPopMatrix();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

- ☐ (0,-2)
- ☐ (2,-2)
- ☐ (-2,2)
- ☐ (-2,-2)

☐ (2,2)

9

The Phong reflectance model does not consider the geometry of the object being shaded
(1 Point)

☒ True

☐ False

10

Does OpenGL use column-major order or row-major order when representing a transformation matrix?
(1 Point)

☐ row-major

☒ column-major

11

OpenGL maintains a set of error flags. Each flag represents a different type of error. Whenever one of these errors occurs, the corresponding flag is set. To see whether any of these flags is set, you should call glGetError
(1 Point)

☒ True

☐ False

12

The process of removal of hidden surfaces is termed as

(1 Point)

- ☐ clipping
- ☐ copying
- ☐ culling
- ☐ shorting

13

glShaderSource() hands shader source code to the shader object so that it can keep a copy of it

(1 Point)

- ☐ True
- ☐ False

14

What does glMaterialf(...) do?

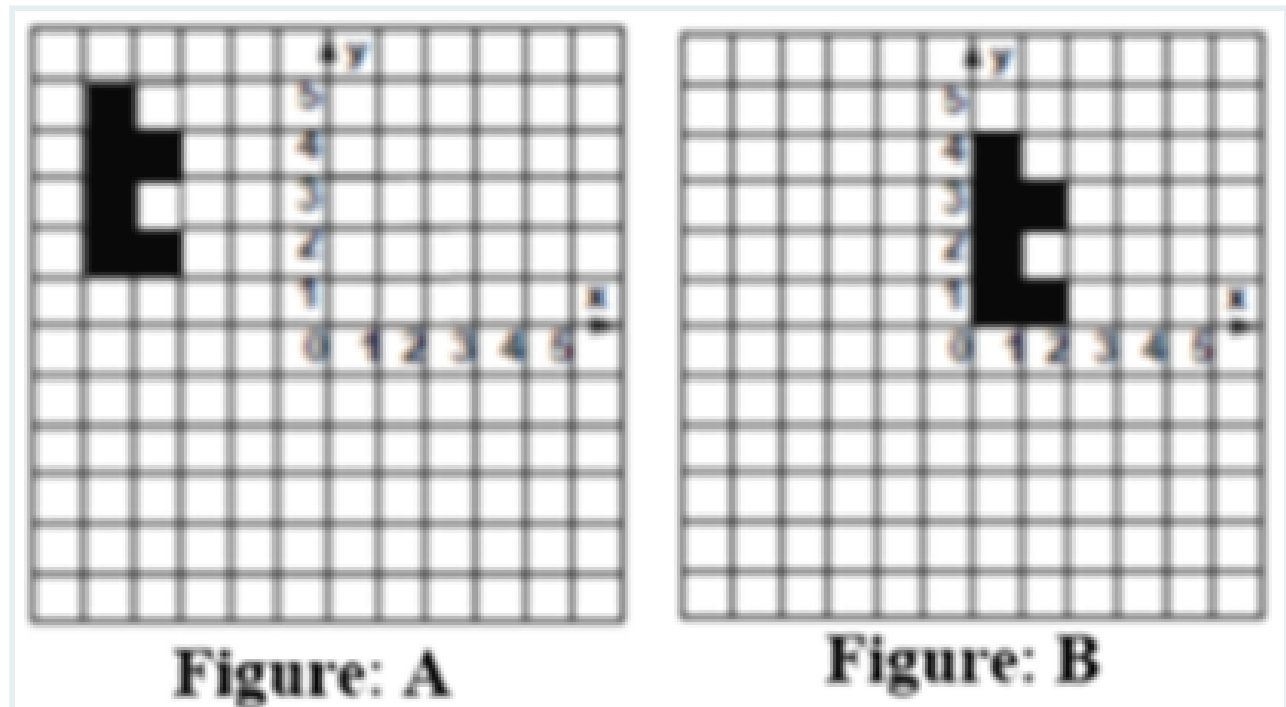
(1 Point)

- ☐ Turns material properties on
- ☐ Switches to material mode to add visual effects
- ☐ Gets the current material properties of a drawn object
- ☐ Defines the material properties of a drawn object for visual effects

15

The OpenGL command(s) that makes the transformation of Figure A to Figure B:

(1 Point)



- ☐ `glTranslatef(5.0, -1.0, 0.0);`
- ☐ `glTranslatef(-5.0, 1.0, 0.0);`
- ☐ `glRotatef(90.0, 0.0, 0.0, 1.0); glTranslatef(-1.0, 1.0, 0.0);`
- ☐ `glTranslatef(-1.0, 1.0, 0.0); glRotatef(90.0, 0.0, 0.0, 1.0);`

16

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glScalef(-2.0, 2.0, 1.0);
```

```
glRotatef(90.0, 0, 0, 1.0);
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☐ C

☒ D

17

OpenGL can draw convex polygons.

(1 Point)

☒ True

☐ False

18

The transformation matrix that corresponds to the following OpenGL code segment is:

```
glRotatef(90.0, 1.0, 0.0, 0.0);
glTranslatef(-2.0,1.0,-3.0);
glScalef(1.0, 2.0,2.0);
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & 2 & 3 \\ 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & -2 & 3 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & -2 & 1 \\ 0 & 2 & 0 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & -2 \\ 0 & 0 & 2 & 3 \\ 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☐ C
- ☐ D

19

If the GL_SPOT_CUTOFF is 180, spotlight effect equals

(1 Point)

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3

20

For a rotation matrix R and a translation matrix T , $RT = TR$, even though matrix multiplication is not generally commutative

(1 Point)

☐ True

☒ False

21

In OpenGL, the argument of _____ determines which object is to draw.

(1 Point)

☐ glEnd()

☒ glBegin()

☐ glStart()

☐ None of these

22

How do you activate a shader program in OpenGL?

(1 Point)

- ☐ glUseProgram(ProgramID);
- ☐ glUseProgram(ProgramID);
- ☐ glProgram(ProgramID);
- ☐ glShader(ProgramID);
- ☐ glUseShader(ProgramID);

23

$$\begin{bmatrix} 0 & -1 & 0 & 4 \\ 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The sequence of OpenGL transformations that would produce the following transformation matrix is:

(1 Point)

- ☐ glRotatef(90, 0, 0, 1); glTranslatef(4,-3,2);
- ☒ glTranslatef(4,-3,2); glRotatef(90, 0, 0, 1);
- ☐ glRotatef(90, 0, 0, 1); glScalef(-1.0, 1.0,1.0); glTranslatef(4,-3,2);
- ☐ glTranslatef(4,-3,2); glScalef(-1.0, 1.0,1.0); glRotatef(90, 0, 0, 1);

24

Suppose that you have 10 triangles that can be arranged into a triangle strip. How many vertices would you have to specify to OpenGL if you use the fact that they can be arranged into a triangle strip?

(1 Point)

- ☐ 9
- ☐ 10
- ☐ 11
- ☒ 12
- ☐ 13

25

Generally, what primitive polygon is used for creating a mesh to represent a complex object?

(1 Point)

- ☒ Triangle
- ☐ Cube
- ☐ Sphere
- ☐ Circle

26

Which of the following equation is used in 2D translation to move a point(x,y) to the new point (x',y')?

(1 Point)

- ☐ $x' = x + ty$ and $y' = y + tx$
- ☐ $x' = x * tx$ and $y' = y * ty$
- ☒ $x' = x + tx$ and $y' = y + ty$
- ☐ $x' = x + tx$ and $y' = y - ty$

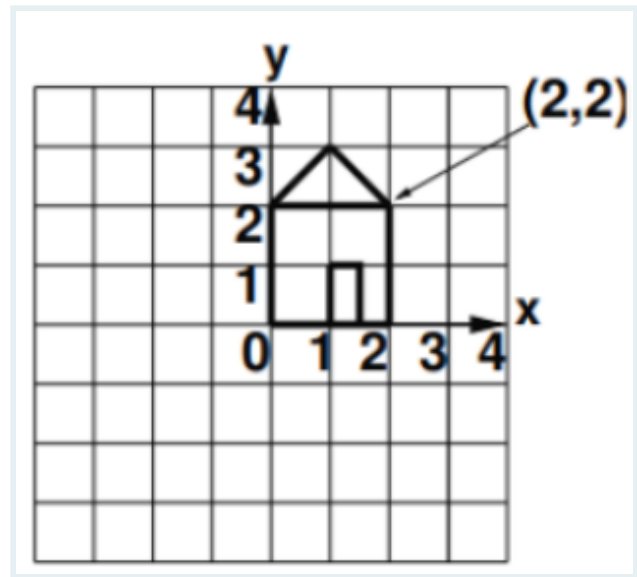
27

The RGB of A cyan color is

(1 Point)

- ☐ (1.0,1.0,0.0)
- ☐ (1.0,0.0,1.0)
- ☒ (0.0,1.0,1.0)
- ☐ (1.0,1.0,1.0)

28



Assume that drawHouse function draws the following house

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glRotate(-90, 0, 0, 1);
drawHouse();
glPopMatrix();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

- ☐ (0,-2)
- ☒ (2,-2)
- ☐ (-2,2)
- ☐ (-2,-2)
- ☐ (2,2)

29

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
glRotatef(90.0, 0, 0, 1.0);
(1 Point)
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☐ C
- ☒ D

30

The fragment shader determines the position of the vertices
(1 Point)

- ☐ True
- ☐ False

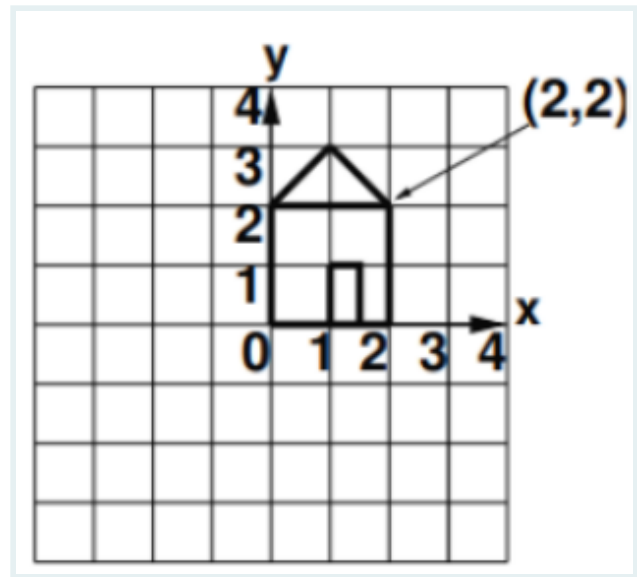
31

What are two types of shaders?

(1 Point)

- ☐ Vertex and polygon.
- ☐ Fragment and polygon.
- ☐ Vertex and fragment.

32



```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
```

```
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef(-1.0, 1.0,1.0);
glRotate(90, 0, 0, 1);
glPopMatrix();
drawHouse();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

☐ (0,-2)

☐ (2,-2)

☐ (-2,2)

☐ (-2,-2)

☒ (2,2)

33

`diffuse_intensity = max(normal dot light_dir, 0)`

(1 Point)

- ☐ True
- ☐ False

34

`glNormal` can be called between `glBegin` and `glEnd`

(1 Point)

- ☒ True
- ☐ False

35

The projection command preserves straight lines

(1 Point)

- ☐ `gluPerspective`
- ☐ `gluLookat`
- ☒ `glOrtho`
- ☐ `glFrustum`
- ☐ `glViewport`

36

If the light position is $(1, 0, 0, 0)$, the attenuation factor at vertex $(0.5, 0.5, 1) =$

.....

(1 Point)

- ☐ Zero
- ☐ 0.5
- ☐ 1
- ☐ Cannot be calculated

37

In the Phong reflectance model, if only diffusion light component is used; a flat surface illuminated with a directional light source will have a uniform color intensity across the surface

(1 Point)

- ☐ True
- ☐ False

38

In OpenGL, lines color is set using____
(1 Point)

- ☒ glColor3f()
- ☐ glColor()
- ☐ glLineColor()
- ☐ None of these

39

What is the significance of i,f,v,d... in function naming conventions found in OpenGL?
(1 Point)

- ☐ OpenGL's naming conventions vary and must be examined in the specification for the version you are using
- ☒ Specify the expected parameter types (int, float, pointer, double, etc.)
- ☐ Specify the target type for variables written by OpenGL on the CPU-side only
- ☐ Specify the output target (immediate mode, frame buffer, virtual mode, double-buffer, etc.)

40

The inverse of a rotation matrix is another matrix such that when the two are multiplied together to get the_____.

(1 Point)

- ☐ Transpose the matrix
- ☒ Identity matrix
- ☐ Diagonal matrix
- ☐ None of these

41

If the rotation matrix R rotates a point by the angle θ , then the transpose of R rotates the point by $-\theta$

(1 Point)

- ☐ True
- ☐ False

42

`glTexCoord*()` is used for placing a text in a specific location in the window

(1 Point)

- ☐ True
- ☐ False

43

The process of displaying 3D into a 2D display unit is called as____
(1 Point)

- ☐ Resolution
- ☒ Projection
- ☐ Rasterization
- ☐ Transformation

44

Once lighting is enabled, what happens to glColor() calls?
(1 Point)

- ☐ A run-time error occurs if both are used.
- ☒ They are ignored.
- ☐ They function as normal.
- ☐ They enhance the shading effects.

45

An example of positional light source is the sun
(1 Point)

- ☐ True
- ☐ False

46

Is the following code, correct?

```
glBegin (LINES);  
glVertex2i (1, 0);  
glVertex2i (1, 1);  
glVertex2i (-1, 1);  
glVertex2i (-1, 0);  
glEnd;
```

(1 Point)

☐ Yes

☒ No

47

Rotating the 3D point (-1,-1,0) by 180 around z-axis gives what:

(1 Point)

☐ (1,-1,0)

☐ (-1,1,0)

☒ (1,1,0)

☐ (0,-1,0)

☐ (-1,-1,0)

48

The plane is given by $2x+2y+z = 6$. What is the normal to the plane N?
(1 Point)

- ☐ (1, 1, 0.5)
- ☐ (2, 2, 3)
- ☒ (2, 1, 0)
- ☐ (2, 0, 1)

49

When using the Phong Reflectance Model, we calculate the red, green, and blue color channels independently
(1 Point)

- ☒ True
- ☐ False

50

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef(-2.0, 2.0, 1.0);
glPushMatrix();
glTranslatef(-1.0f, -2.0f, -3.0f);
glRotatef(-90.0, 0, 0, 1.0);
glScalef (-1.0,0.5, -0.5);
glPopMatrix();
glRotatef(90.0, 0, 0, 1.0);
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -0.5 & 0 & 0 \\ 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 & -1.0 \\ 0 & -2 & 0 & -2.0 \\ 0 & 0 & 1 & -3.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☐ C
- ☒ D

51

OpenGL has no functions for keyboard input or mouse interaction

(1 Point)

- ☒ True
- ☐ False

52

..... light component comes from one direction and once it hits a surface, it is scattered.

(1 Point)

- ☒ Diffuse
- ☐ Specular
- ☐ Ambient
- ☐ Spot

53

Write down the Modelview matrix after executing the following OpenGL code segment.

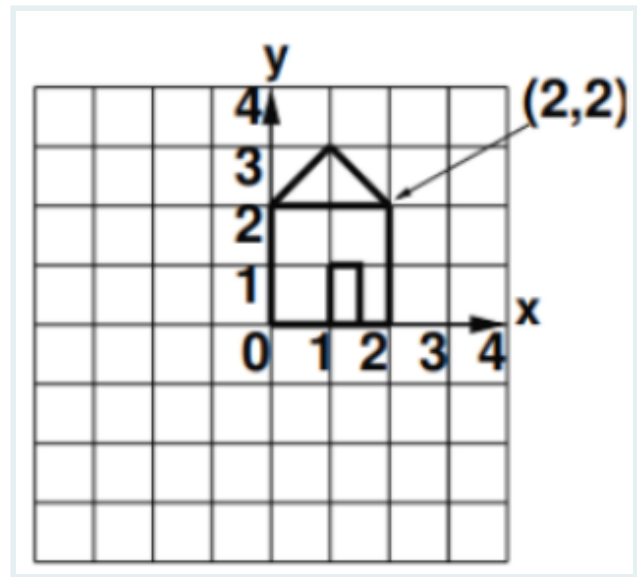
```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef (-1.0,1.0, -1.0);
glScalef (1.0,0.5, 1.0);
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0.5 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☒ C
- ☐ D

54



Assume that drawHouse function draws the following house

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef(1.0, -1.0,1.0);
drawHouse();
glPopMatrix();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

- ☐ (0,-2)
- ☒ (2,-2)
- ☐ (-2,2)
- ☐ (-2,-2)
- ☐ (2,2)

55

In glVertex2i() command, 2 represents _____
(1 Point)

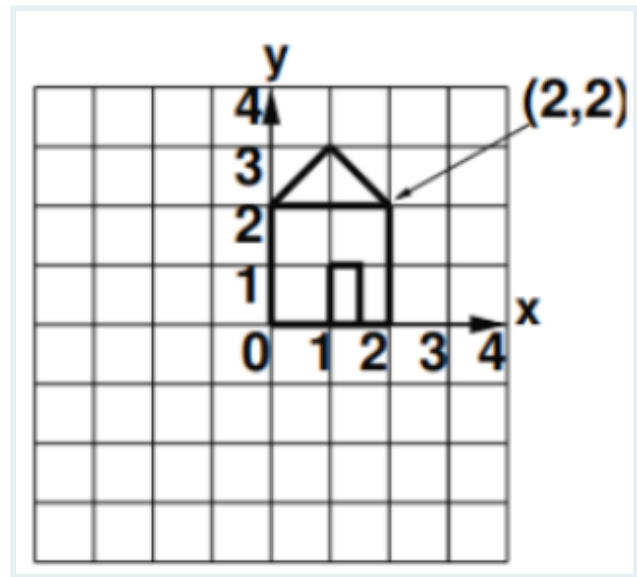
- ☐ type of arguments
- ☐ basic command
- ☐ number of arguments
- ☐ None of these

56

diffusion light component is view independent
(1 Point)

- ☐ True
- ☐ False

57



```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
```

```
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0); glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef(-1.0, 1.0,1.0);
glRotate(90, 0, 0, 1);
drawHouse();
glPopMatrix();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:

(1 Point)

- ☐ (0,-2)
- ☐ (2,-2)
- ☐ (-2,2)
- ☐ (-2,-2)
- ☒ (2,2)

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